

UNIVERSITÀ DEGLI STUDI DI MILANO-BICOCCA

DATA SCIENCE LAB FOR SMART CITIES

FINAL ESSAY

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# The 15-Minute City: Measuring Service Accessibility Inequality Across Neighborhoods in Barcelona

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## Abstract

This essay examines service accessibility inequality in Barcelona through the framework of the 15-minute city. The study was developed to assess whether essential urban services are equally accessible across neighbourhoods, and to understand how proximity-based planning can support more inclusive smart city policies. Using open municipal data, OpenStreetMap data, and geospatial analysis in Python, a neighbourhood-level accessibility index is constructed for healthcare facilities, schools, supermarkets, pharmacies, parks, and metro stations.

The essay combines spatial analysis with socioeconomic indicators to evaluate the relationship between service accessibility, income, and population distribution. It also applies clustering methods to identify different neighbourhood profiles and to support policy-oriented interpretation. The significance of the study lies in showing how open data and reproducible data science methods can help urban decision makers identify underserved areas, evaluate spatial inequality, and design more equitable 15-minute city strategies.

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# 1 Introduction and Problem Description

## 1.1 The 15-Minute City Concept

One of the most important urban planning ideas of the twenty-first century is the "15-minute city". First systematically theorized by French-Colombian urbanist Carlos Moreno and subsequently adopted as official policy by the city of Paris in 2020, the concept proposes a simple principle: every urban resident should be able to reach all essential daily services - schools, healthcare facilities, food retail, pharmacies, green spaces, and public transport - within fifteen minutes on foot or by bicycle from their home [1]. This framework is relevant not only for its environmental advantages, such as decreased reliance on automobiles, lower carbon emissions, and better public health outcomes, but also for its implicit promise of social equity: a city structured around proximity is one that, in theory, functions equally well for all of its citizens, regardless of their age, income, or mobility.

The concept gained renewed importance during the COVID-19 pandemic, when lockdown measures forced many urban residents to rely on their immediate surroundings. This experience made it clear that local services were not distributed evenly throughout urban areas. The majority of their everyday demands might be satisfied with a short walk for residents of crowded, mixed-use central neighbourhoods. On the other hand, people who lived in peripheral or monofunctional neighbourhoods often had to take public transport or make longer trips in order to get the same services. Because of this, the 15-minute city raises both spatial justice and urban design issues.

## 1.2 Barcelona as a Case Study

Barcelona is an excellent case study for researching the 15-minute city concept for three interconnected reasons. First, Barcelona's Superblock initiative, or Superilla in Catalan, has made a clear commitment to this urban paradigm. The Superblock plan, which was started under former mayor Ada Colau and carried on under succeeding administrations, reorganises city blocks by limiting through traffic, making space for bicyclists and pedestrians, and promoting the installation of local services. This gives our findings immediate policy importance because Barcelona is now an active participant in the 15-minute city transition rather than just a passive subject of analysis.

Second, there are significant spatial disparities throughout Barcelona. From some of the wealthiest residential areas in Spain, like Pedralbes and les Tres Torres in the Sarrià-Sant Gervasi district, where the average household disposable income exceeds 45,000 euros annually, to some of Catalonia's most economically deprived urban neighbourhoods, like Ciutat Meridiana, Torre Baró, and Vallbona in the Nou Barris district, where the average household income falls below 15,000 euros annually [2]. The analytical conditions required to determine whether service accessibility follows, contradicts, or functions independently of socioeconomic geography are created by this income gap of more than three to one between the wealthiest and poorest neighbourhoods.

Third, Barcelona has one of Europe's largest and easily accessible municipal open data ecosystems. In order to facilitate rigorous quantitative analysis without depending on proprietary or commercially restricted data sources, the Ajuntament de Barcelona's Open Data BCN portal regularly releases updated datasets on population, household income, public services, and administrative boundaries at the neighbourhood level.

### 1.3 Research Question and Hypothesis

This study addresses the following central research question:

Is Barcelona equally a 15-minute city for all its residents, or does service accessibility vary across neighbourhoods in ways that reflect broader spatial and socioeconomic inequalities?

The initial hypothesis was that higher-income neighbourhoods would have better access to essential services, in line with the broader literature on accessibility, equity, and urban inequality [3, 4]. However, the analysis also tests whether income alone explains service accessibility, or whether other urban factors, such as geographic centrality, density, urban morphology, and mixed land use, play a stronger role. This distinction is important because a simple rich-versus-poor explanation may overlook the complexity of Barcelona’s urban structure.

### 1.4 Significance and Contribution

The significance of this study lies in combining spatial analysis and socioeconomic data to produce a neighbourhood-level accessibility score for all seventy-three neighbourhoods of Barcelona. The score is based on the number of essential services reachable within an 800-metre straight-line radius from each neighbourhood centroid. This distance is used as an approximation of a ten-to-fifteen-minute walk.

By combining the accessibility score with income and population data, the analysis identifies different types of neighbourhoods: areas that are well served, areas that are underserved, areas that are well served despite lower income, and areas that are poorly served despite higher income. The resulting cluster typology provides a practical tool for policy prioritization and contributes to the broader discussion on how smart city strategies can support spatial equity rather than reinforce existing inequalities.

## 2 Study Area, Data and Methodology

### 2.1 Study Area

Barcelona is the second-biggest city in Spain and the capital of Catalonia. Situated on the northeastern Mediterranean coast, the city is home to nearly 1.6 million people and has an area of roughly 101 square kilometres. The neighbourhood, referred to as the *barri* in Catalan, serves as the study’s spatial unit of analysis. Ten districts, which are further subdivided into seventy-three neighbourhoods, make up Barcelona’s administrative divisions.

The entire spatial foundation of the analysis is made up of the seventy-three neighbourhoods. Since no neighbourhoods were left out, the study was able to fully reflect the range of urban circumstances in Barcelona, from peripheral, hillside, and lower-density districts to congested center neighbourhoods. This is significant since the 15-minute city model heavily relies on the spatial distribution of services, urban density, and shape.

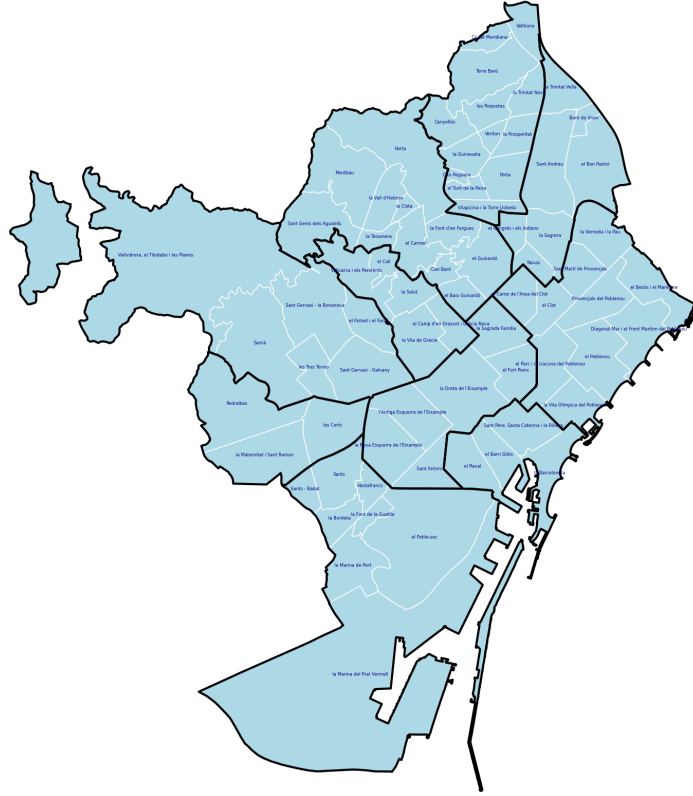


Figure 1: Barcelona: neighbourhoods and district boundaries.

Figure 1 presents the base map of Barcelona, showing the seventy-three neighbourhoods and the ten district boundaries. This map provides the geographical reference for the accessibility analysis developed in the following sections.

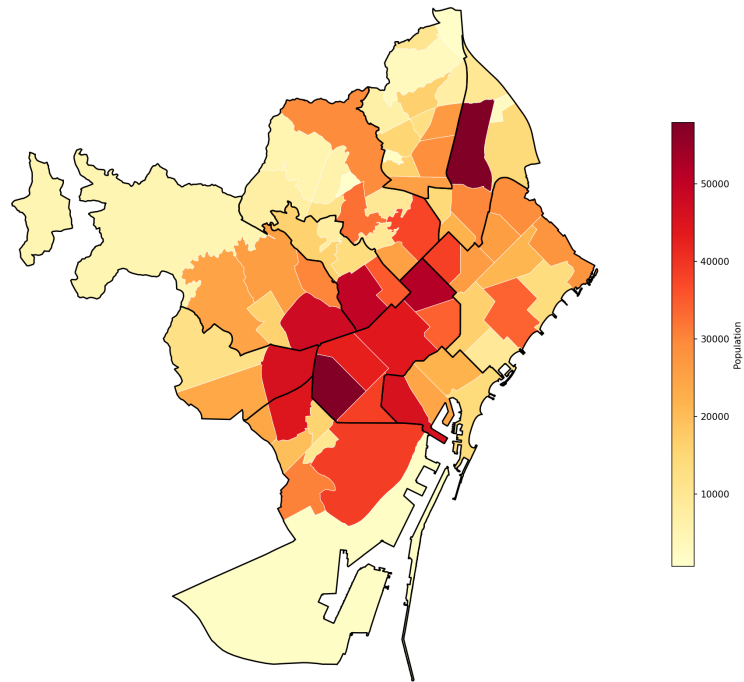


Figure 2: Barcelona: population by neighbourhood.

Population is also unevenly distributed across Barcelona. Some central and inner-city neighbourhoods contain high numbers of residents, while peripheral or hillside neighbourhoods have much lower population levels. This matters because accessibility gaps affect not only places, but also the number of residents living in those places.

Barcelona also presents a clear spatial pattern of socioeconomic inequality. Higher-income neighbourhoods are mainly concentrated in the western and north-western parts of the city, particularly in Sarrià-Sant Gervasi and Les Corts. In contrast, lower-income neighbourhoods are more visible in parts of Nou Barris, Sant Andreu, and peripheral areas of the city.

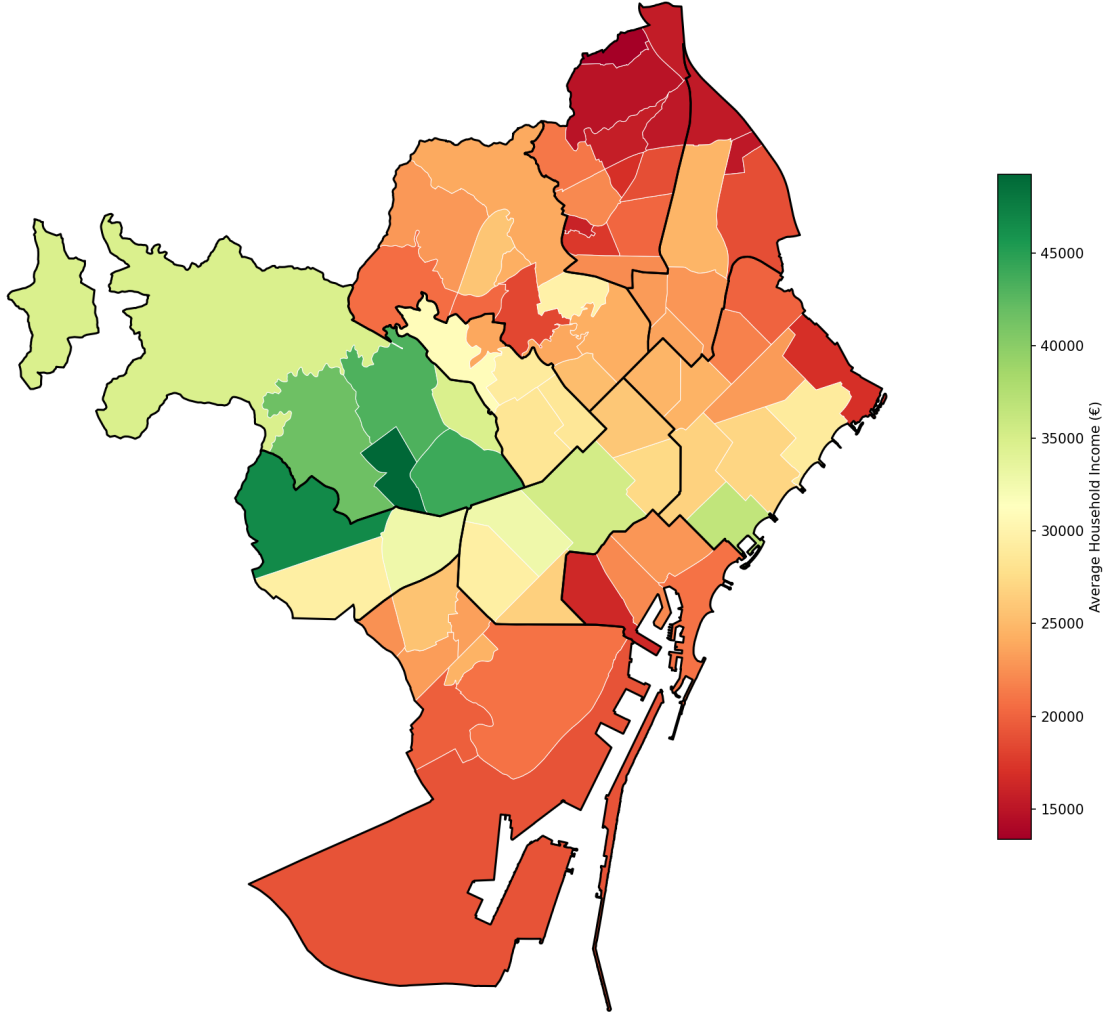


Figure 3: Barcelona: average household income by neighbourhood.

Figure 3 provides the socioeconomic background for the study. It allows the accessibility results to be interpreted not only as a spatial pattern, but also as a question of social equity.

## 2.2 Data Sources

This study combines municipal open data with geospatial data extracted from OpenStreetMap. The datasets include administrative boundaries, population, income, healthcare facilities, essential services, and the pedestrian street network. Table 1 summarizes the main datasets used in the analysis and provides the corresponding download sources.

Table 1: Data sources used in the analysis

| Dataset                                                  | Source                                   | Format                     | Role in the analysis                                                                                     |
|----------------------------------------------------------|------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------|
| Neighbourhood boundaries                                 | <code>geometries_cat_bcn_2024</code> [5] | GeoJSON                    | Spatial base layer for the 73 neighbourhoods. The file <code>barris.geojson</code> was used.             |
| District boundaries                                      | <code>geometries_cat_bcn_2024</code> [5] | GeoJSON                    | District overlay used for spatial reference. The file <code>districtes.geojson</code> was used.          |
| Population data                                          | Open Data BCN [6]                        | CSV                        | Population by neighbourhood, used to estimate how many residents are affected by poor accessibility.     |
| Household income data                                    | Open Data BCN [2]                        | CSV                        | Socioeconomic indicator used to compare accessibility with household income.                             |
| Healthcare facilities                                    | Open Data BCN [7]                        | CSV                        | Official dataset of hospitals and primary health centres used for healthcare accessibility.              |
| Schools, supermarkets, pharmacies, parks, metro stations | OpenStreetMap OSMnx [8, 9]               | via GeoDataFrame / GeoJSON | Essential service locations used to calculate the accessibility score.                                   |
| Walking street network                                   | OpenStreetMap OSMnx [8, 9]               | via GraphML                | Pedestrian network dataset downloaded for mobility context and possible future network-based extensions. |

The neighbourhood and district boundary files were obtained from the `geometries_cat_bcn_2024` GitHub repository. The file `barris.geojson` was used for neighbourhood boundaries, while `districtes.geojson` was used for district boundaries. These files form the spatial base layer of the project because every map and neighbourhood-level calculation depends on them.

Population data were obtained from the Open Data BCN dataset Population of Barcelona according to the Municipal Register of Inhabitants [6]. This dataset was used to calculate how many residents may be affected by weak accessibility in each neighbourhood.

Household income data were obtained from the Open Data BCN dataset Disposable Household Income per Capita in the City of Barcelona [2]. This dataset provides the inequality dimension of the analysis, making it possible to compare accessibility patterns with socioeconomic differences between neighbourhoods.

Healthcare facilities were obtained from the Open Data BCN dataset Hospitals and Primary Health Services of the City of Barcelona [7]. This official municipal dataset was used for healthcare accessibility because it provides more reliable healthcare facility information than relying on OpenStreetMap alone.

Service location data for schools, supermarkets, pharmacies, parks, and metro stations were extracted from OpenStreetMap using the OSMnx Python library [8, 9]. The final service dataset used in the accessibility analysis included 510 schools, 1,068 supermarkets, 1,024 pharmacies, 622 parks, 143 metro stations, and 146 healthcare facilities.

The walking street network for Barcelona was also downloaded using OSMnx. The downloaded network contained 73,278 nodes and 233,638 edges, representing pedestrian-accessible intersections and street segments. Although this network was not used directly in the final accessibility score, it provides useful mobility context and can support future extensions of the analysis based on network walking distance instead of straight-line distance.

## 2.3 Methodology

The analytical methodology consists of four main steps: data preparation and cleaning, accessibility score calculation, inequality analysis, and neighbourhood clustering.

### 2.3.1 Data Preparation and Cleaning

All spatial datasets were processed in Python using GeoPandas. The neighbourhood and district boundary files were loaded as GeoDataFrames and initially stored in the WGS84 geographic coordinate system, EPSG:4326, which is suitable for mapping and visualization. For distance-based calculations, all spatial datasets were reprojected to EPSG:25831, a projected coordinate reference system appropriate for Barcelona because it expresses distances in metres.

Population data were aggregated to the neighbourhood level using grouped summation, producing one total population value for each of the seventy-three neighbourhoods. Income data were aggregated using grouped averaging, producing one average income value for each neighbourhood. This made it possible to merge both socioeconomic indicators with the neighbourhood boundary layer.

Healthcare data required additional preprocessing because the CSV file was encoded in UTF-16. After resolving the encoding issue, the latitude and longitude columns were used to convert the dataset into a GeoDataFrame. Service datasets extracted from OpenStreetMap included different geometry types, such as points and polygons. For the accessibility analysis, these geometries were represented as point locations using centroid calculation after reprojection to the metric coordinate system.

### 2.3.2 Accessibility Score Calculation

The main analytical indicator developed in this study is a composite accessibility score for each neighbourhood. The score measures how many essential services are located within an 800-metre straight-line buffer from the geographic centroid of each neighbourhood. The 800-metre threshold is used as an approximation of a ten-to-fifteen-minute walking distance. This is not a network-based walking-time isochrone, but a proximity-based measure that allows comparable analysis across all neighbourhoods.

For each neighbourhood, the following service categories were counted within the 800-metre buffer: healthcare facilities, schools, supermarkets, pharmacies, parks, and metro stations. These counts were then combined into a weighted composite score:

$$\begin{aligned}\text{Accessibility Score} = & (3.0 \times \text{Healthcare}) + (2.5 \times \text{Metro Stations}) \\ & + (2.0 \times \text{Schools}) + (2.0 \times \text{Supermarkets}) \\ & + (2.0 \times \text{Pharmacies}) + (1.0 \times \text{Parks})\end{aligned}$$

The weights reflect the relative importance of each service category for everyday urban life. Healthcare facilities received the highest weight because access to medical services is essential, especially for elderly residents, low-income households, and people with reduced mobility. Metro stations were also weighted highly because public transport access connects residents to employment, education, and services beyond their immediate neighbourhood. Schools, supermarkets, and pharmacies were assigned intermediate weights because they represent essential daily needs. Parks were included as an important quality-of-life indicator, but with a lower weight than healthcare, food retail, and transport.

The raw accessibility scores were normalized to a scale from 0 to 100 using min-max normalization:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \times 100$$

In this scale, 100 represents the neighbourhood with the highest relative accessibility score in Barcelona, while 0 represents the neighbourhood with the lowest relative accessibility score. Therefore, the score should be interpreted as a relative index comparing Barcelona neighbourhoods with each other, not as an absolute measure of universal service accessibility.

### 2.3.3 Inequality Analysis

To evaluate the relationship between socioeconomic status and service accessibility, the normalized accessibility score was merged with neighbourhood-level income and population data. The Pearson correlation coefficient was calculated between average household income and the normalized accessibility score across all seventy-three neighbourhoods.

A scatter plot was also produced to visualize this relationship. In the plot, the horizontal axis represents average household income, the vertical axis represents the accessibility score, and the size of each point represents neighbourhood population. This allows the analysis to consider three dimensions simultaneously: socioeconomic status, service accessibility, and the number of residents potentially affected.

### 2.3.4 Neighbourhood Clustering

To produce a policy-oriented typology of neighbourhoods, K-Means clustering was applied using three variables: normalized accessibility score, average income, and population. Before clustering, all variables were standardized using `StandardScaler`, because the three indicators are measured on different numerical scales.

The number of clusters was selected using the elbow method. This method compares the reduction in within-cluster inertia as the number of clusters increases. In this study, the elbow plot suggested that four clusters provided a reasonable balance between interpretability and within-cluster similarity. The final K-Means model was therefore fitted with  $k = 4$ .

The resulting clusters were labelled according to their average accessibility, income, and population characteristics:

- High Access;
- Medium-High Access;
- Medium-Low Access;
- Low Access / Underserved.

These clusters were later mapped to reveal the geographic distribution of accessibility profiles across Barcelona and to support policy interpretation.

## 3 Results and Analysis

### 3.1 Overview of Service Distribution in Barcelona

Before examining accessibility at the neighbourhood level, it is useful to consider the overall distribution of essential services across the city.

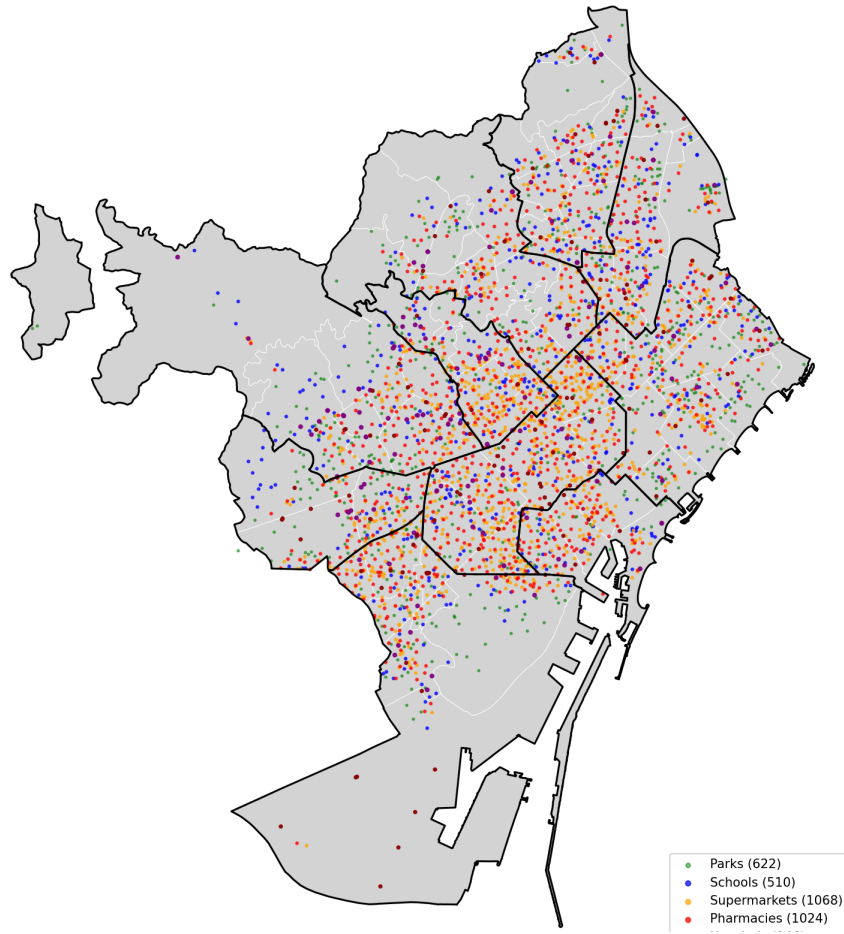


Figure 4: Barcelona: all essential services overview. The analysis includes 622 parks, 510 schools, 1,068 supermarkets, 1,024 pharmacies, 146 healthcare facilities, and 143 metro stations.

Figure 4 presents a comprehensive overview of the 3,513 service locations included in the analysis. These include supermarkets, pharmacies, parks, schools, healthcare facilities, and metro stations.

Several spatial patterns are visible. First, services are strongly concentrated in the central and inner-city districts, especially Eixample, Gracia, Ciutat Vella, and parts of Sant Marti. These areas have a dense urban fabric, mixed land use, and a high concentration of commercial and public facilities. Second, the western districts of Sarria-Sant Gervasi and Les Corts, despite their higher income levels, show lower densities of some daily services compared with the central districts. This reflects their more residential and lower-density urban form. Third, peripheral areas in the north and north-west, including parts of Nou Barris and Horta-Guinardo, show a lower concentration of services. Finally, southern port and industrial areas, such as la Marina del Prat Vermell, contain fewer daily services, reflecting their industrial and logistical land-use structure.

This first visual inspection already suggests that Barcelona's 15-minute city accessibility is not evenly distributed. Service proximity appears to be shaped not only by income, but also by centrality, density, land-use mix, and urban morphology.

### 3.2 Neighbourhood Accessibility Scores

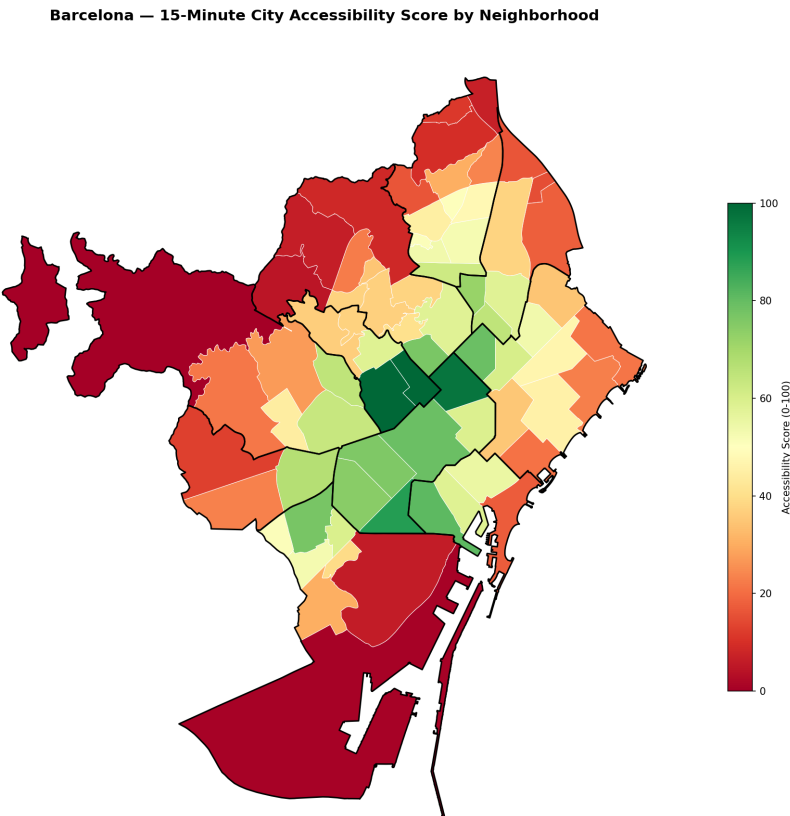


Figure 5: Barcelona: 15-minute city accessibility score by neighbourhood, normalized on a 0–100 scale.

Figure 5 presents the normalized accessibility score for all seventy-three neighbourhoods of Barcelona. The score ranges from 100.00 for la Vila de Gràcia, the most accessible neighbourhood in the analysis, to 0.00 for Vallvidrera, el Tibidabo i les Planes, the least accessible neighbourhood according to the relative index.

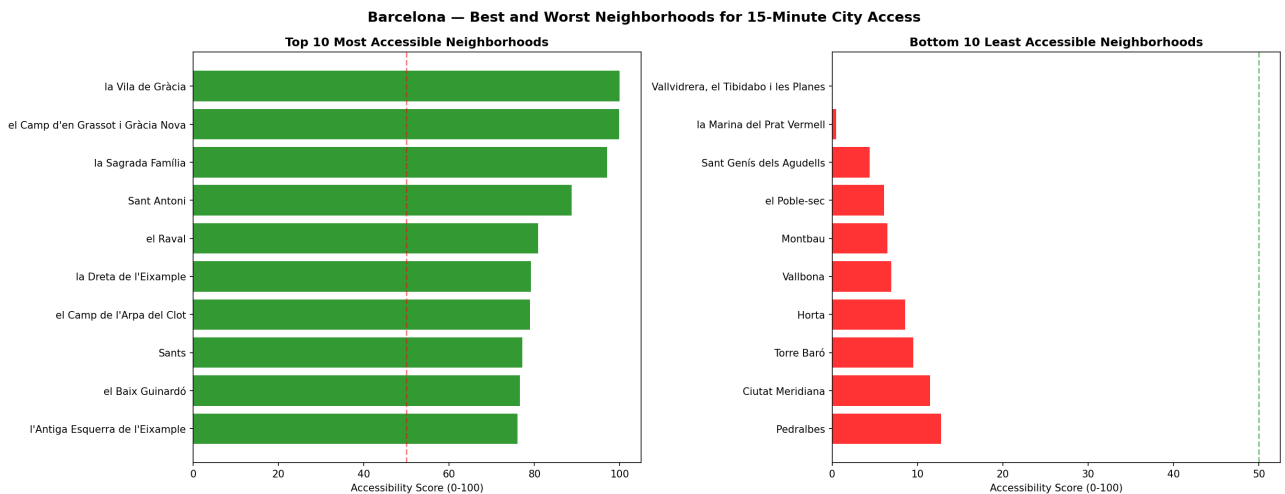


Figure 6: Barcelona: top 10 most accessible and bottom 10 least accessible neighbourhoods.

The accessibility map and ranking in Figures 5 and 6 show a clear centre-periphery pattern. The highest scores are found in dense, mixed-use, and centrally located neighbourhoods such as la Vila de Gracia, la Sagrada Familia, Sant Antoni, el Raval, and la Dreta de l'Eixample. These areas benefit from a high concentration of daily services, compact urban form, and strong public transport coverage.

The lowest scores are mainly found in peripheral, mountainous, low-density, or industrial areas, including Vallvidrera, la Marina del Prat Vermell, Sant Genis dels Agudells, Vallbona, Torre Baro, Ciutat Meridiana, and Pedralbes. El Poble-sec is an exception because it is relatively central, so its low score may partly reflect the centroid-based methodology and the influence of Montjuic.

Two examples highlight the complexity of the results. El Raval has a high accessibility score despite being a lower-income neighbourhood, while Pedralbes has a low accessibility score despite being a high-income area. This shows that accessibility inequality in Barcelona cannot be explained by income alone; urban density, centrality, and land-use structure also play an important role.

### 3.3 Income Versus Accessibility

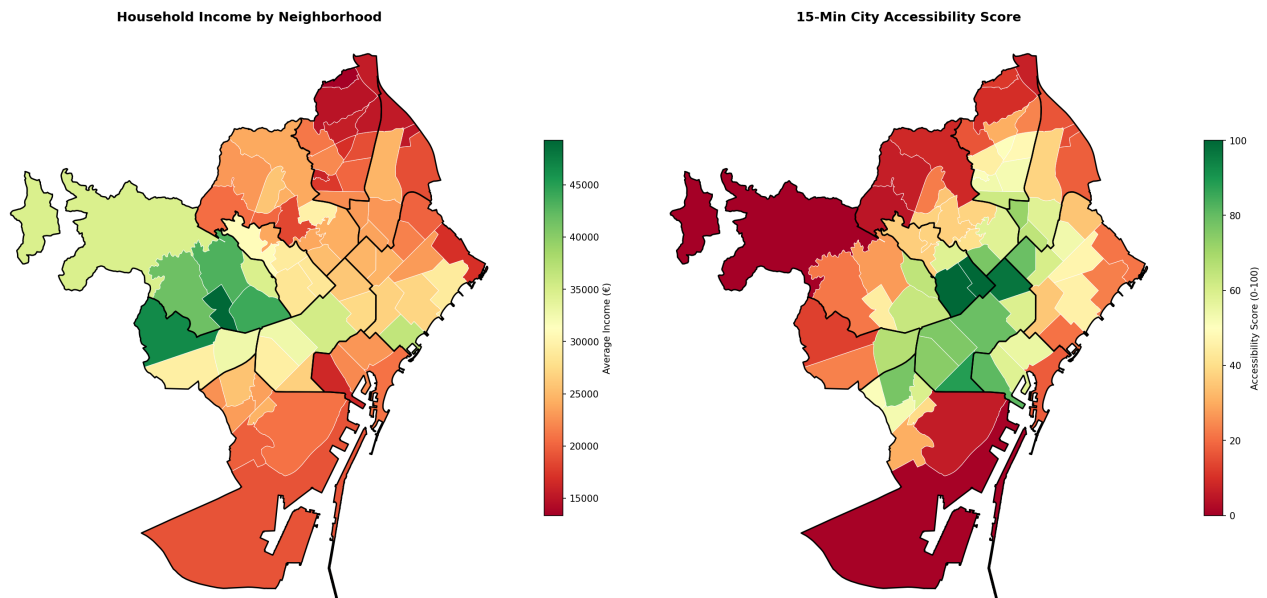


Figure 7: Barcelona: side-by-side comparison of household income and 15-minute city accessibility score by neighbourhood.

Figure 7 compares the spatial distribution of household income with the spatial distribution of accessibility scores. The two maps show that income and accessibility do not follow the same spatial pattern. High-income areas are mainly concentrated in the western and north-western parts of the city, while high-accessibility areas are more strongly concentrated in central and inner-city neighbourhoods.



Figure 8: Barcelona: income versus 15-minute city accessibility score. Bubble size represents neighbourhood population.

Figure 8 presents the relationship between average household income and accessibility score. The Pearson correlation coefficient between income and accessibility is 0.167, indicating a weak positive relationship. This means that higher-income neighbourhoods are only slightly more likely to have higher accessibility scores. In practical terms, income alone is not a strong predictor of service accessibility in Barcelona.

The scatter plot suggests four broad neighbourhood profiles. First, there are low-income but high-accessibility neighbourhoods, such as el Raval, where centrality and density generate strong service proximity despite socioeconomic vulnerability. Second, there are high-income but low-accessibility neighbourhoods, such as Pedralbes and parts of Sarrià-Sant Gervasi, where low-density residential urban form reduces local service proximity. Third, there are moderate-income and high-accessibility neighbourhoods, especially in Gràcia, Eixample, and other central areas. Fourth, there are low-income and low-accessibility neighbourhoods, such as Ciutat Meridiana, Torre Baró, Vallbona, and Baró de Viver.

The most policy-relevant group is the fourth one. These neighbourhoods experience a form of double disadvantage: residents face both lower economic resources and weaker access to essential services within walking distance. This group should be the main priority for equity-oriented 15-minute city policies.

### 3.4 Neighbourhood Clustering Results

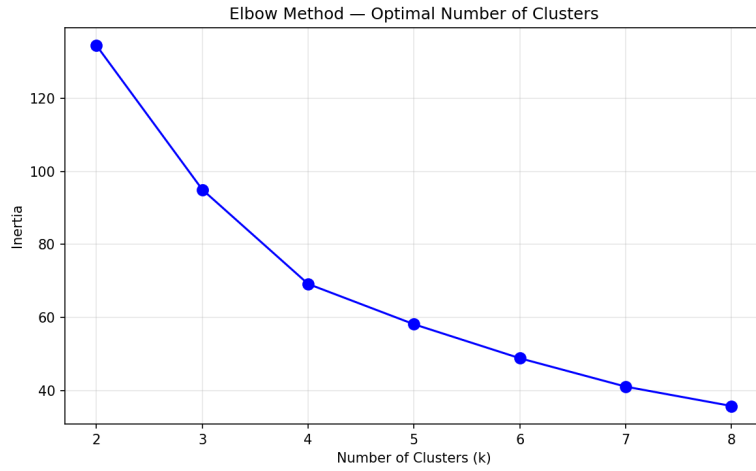


Figure 9: Elbow method for determining the number of clusters. The plot supports the selection of  $k = 4$ .

To produce a policy-oriented typology of neighbourhoods, K-Means clustering was applied using three standardized variables: accessibility score, average income, and population. The elbow plot in Figure 9 suggests that four clusters provide a reasonable balance between interpretability and within-cluster similarity.

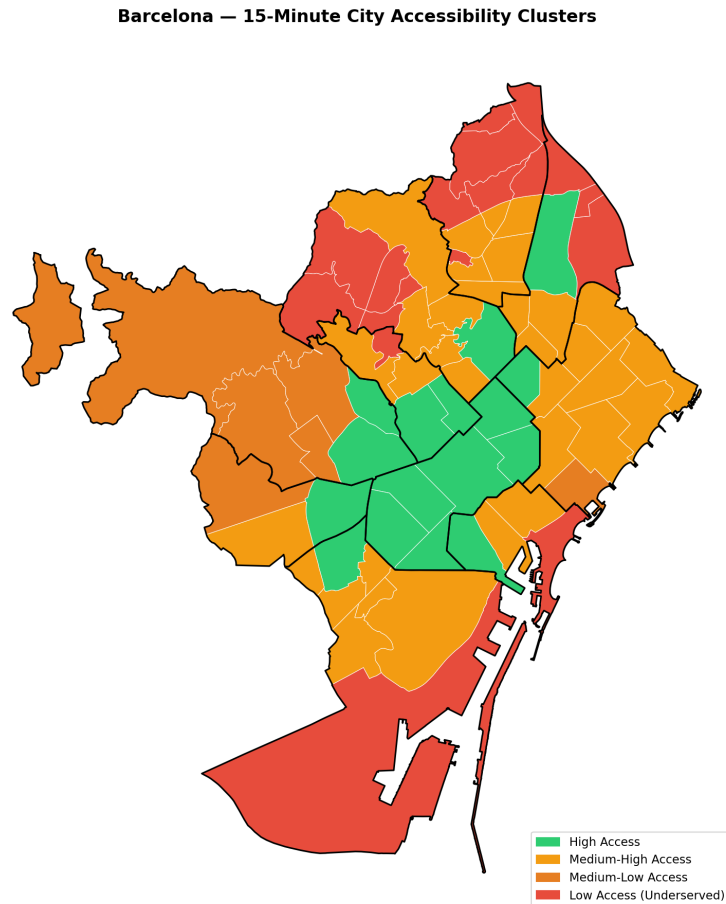


Figure 10: Barcelona: 15-minute city accessibility clusters.

Figure 10 shows the spatial distribution of the four clusters across the city. The clusters should be interpreted as multidimensional neighbourhood profiles, because they are based not only on accessibility, but also on income and population.

Table 2: Cluster characteristics

| Cluster                  | Neighbourhoods | Accessibility | Income (€) | Population |
|--------------------------|----------------|---------------|------------|------------|
| High Access              | 16             | 75.15         | 28,837     | 43,867     |
| Medium-High Access       | 33             | 45.36         | 23,204     | 22,257     |
| Medium-Low Access        | 6              | 21.15         | 42,015     | 15,595     |
| Low Access / Underserved | 18             | 19.91         | 18,820     | 7,252      |

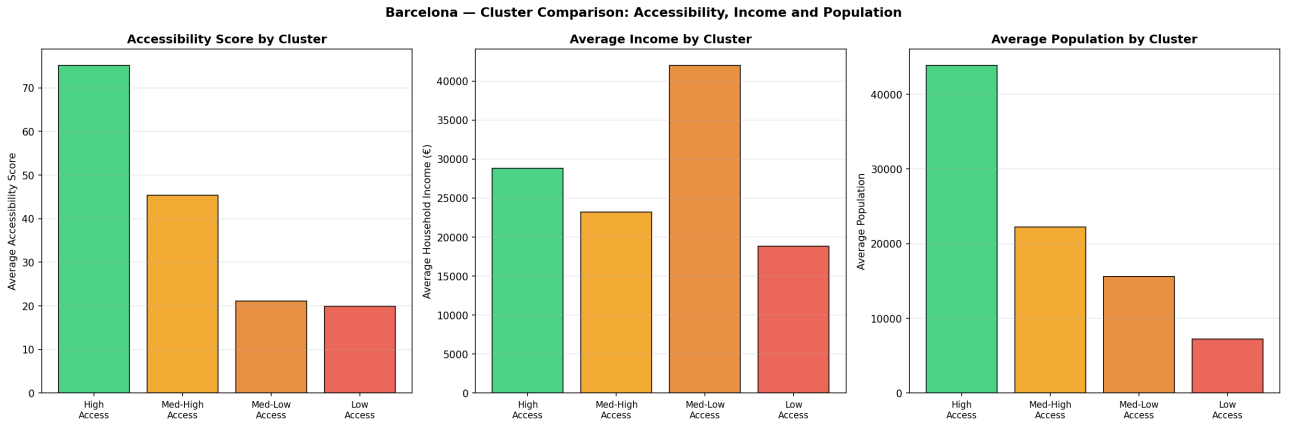


Figure 11: Barcelona: cluster comparison by accessibility score, average income, and average population.

The High Access cluster contains sixteen neighbourhoods with the highest average accessibility score, 75.15. These neighbourhoods include el Raval, la Sagrada Familia, la Dreta de l'Eixample, Sant Antoni, la Vila de Gracia, Sants, les Corts, el Camp de l'Arpa del Clot, and Sant Andreu. They are generally dense, well-connected, and rich in local services. The cluster also has the highest average population, approximately 43,867 residents per neighbourhood, meaning that high accessibility benefits a large number of residents.

The Medium-High Access cluster is the largest group, containing thirty-three neighbourhoods. Its average accessibility score is 45.36 and its average income is approximately 23,204 euros. This group forms a broad middle category of neighbourhoods with moderate service proximity. However, because K-Means clustering uses income and population as well as accessibility, not every neighbourhood in this cluster has medium-high accessibility individually. The cluster should therefore be interpreted as a general neighbourhood profile rather than as a simple ranking.

The Medium-Low Access cluster contains six neighbourhoods: Pedralbes, Vallvidrera, el Tibidabo i les Planes, Sarria, les Tres Torres, Sant Gervasi-la Bonanova, and la Vila Olimpica del Poblenou. This is one of the most analytically interesting groups because it has the highest average income, approximately 42,015 euros, but a low average accessibility score of 21.15. This shows that high income does not necessarily produce high 15-minute city accessibility. In these cases, low accessibility is more closely associated with low-density residential form, peripheral location, or separation from dense commercial streets.

The Low Access / Underserved cluster contains eighteen neighbourhoods and represents the most important group for policy intervention. Its average accessibility score is 19.91 and its average income is approximately 18,820 euros. This cluster includes several neighbourhoods in Nou Barris, Horta-Guinardo, Sant Andreu, and the southern port zone. These neighbourhoods combine weaker service

proximity with lower average income, creating a double disadvantage. Based on the average population of the cluster, this group represents approximately 130,000 residents.

The spatial distribution of clusters reinforces the main finding of the study. Barcelona’s 15-minute city accessibility is strongest in the dense central and inner-city areas, while lower accessibility is concentrated in peripheral, hillside, industrial, or lower-density neighbourhoods. However, the relationship with income is not linear. Some low-income central neighbourhoods have high accessibility, while some high-income residential neighbourhoods have low accessibility. The main equity concern is therefore not simply income inequality, but the overlap between low income and weak local service proximity.

## 4 Policy Recommendations

### 4.1 Overview

The findings presented in Section 3 show that Barcelona’s ambition to become a 15-minute city is not uniformly realized across its seventy-three neighbourhoods. Dense and central areas already perform relatively well, while the Low Access / Underserved cluster contains eighteen neighbourhoods with weaker proximity to essential services and lower average household income. Based on the average population of this cluster, approximately 130,000 residents are affected by this double disadvantage.

This section translates the analytical findings into six policy recommendations for Barcelona’s urban planning and smart city governance. The recommendations are based on the accessibility map in Figure 5, the cluster map in Figure 10, the income-accessibility comparison in Figure 7, and the cluster comparison in Figure 11.

### 4.2 Recommendation 1: Prioritize Public Investment in Low-Access and Low-Income Neighbourhoods

The first priority should be neighbourhoods where low accessibility overlaps with lower average income. The Low Access / Underserved cluster includes several neighbourhoods in Nou Barris, such as Can Peguera, Canyelles, les Roquetes, la Trinitat Nova, Torre Baro, Ciutat Meridiana, and Vallbona. Other low-access neighbourhoods are located in Horta-Guinardo, Sant Andreu, and the southern port zone.

These areas should be treated as priority zones for equity-oriented 15-minute city investment. Public investment should focus on essential services that directly affect everyday life, including healthcare facilities, schools, food retail, pharmacies, public space, and safe pedestrian connections.

The Ajuntament de Barcelona should therefore designate the Low Access / Underserved cluster as a priority intervention area. Investment should be coordinated with existing neighbourhood regeneration programmes, such as the *Pla de Barris*, and should include measurable accessibility targets. For example, future planning documents could aim to increase the accessibility score of the lowest-performing neighbourhoods within a defined period.

### 4.3 Recommendation 2: Improve Public Transport and Last-Mile Connectivity

Public transport is particularly important in peripheral areas because it works as an accessibility multiplier. Even when services are not available locally, reliable public transport can connect residents to employment, education, healthcare, and other facilities across the city.

The results show that several low-access neighbourhoods have weak proximity to metro stations and essential services within the 800-metre buffer. However, the analysis does not prove that one specific metro extension is the best solution. Therefore, the recommendation is to improve public transport and last-mile connectivity through a combination of short-term and long-term measures.

In the short term, high-frequency bus connections should be reinforced between underserved neighbourhoods and existing metro or railway stations. This is especially relevant for hillside and peripheral

areas where metro construction may be technically difficult or costly. Pedestrian access to public transport should also be improved through safer crossings, better lighting, accessible sidewalks, and clearer wayfinding.

In the medium and long term, future metro, tram, or bus rapid transit extensions should use accessibility indicators as part of station-placement decisions. Transport planning should consider not only expected demand, but also whether new connections reduce spatial inequality.

#### **4.4 Recommendation 3: Use Planning Incentives in Wealthy but Low-Access Neighbourhoods**

The Medium-Low Access cluster shows that low accessibility is not always linked to poverty. This cluster has the highest average income, approximately 42,015 euros, but a low average accessibility score of 21.15. It includes neighbourhoods such as Pedralbes, Sarria, les Tres Torres, Sant Gervasi-la Bonanova, Vallvidrera, el Tibidabo i les Planes, and la Vila Olímpica del Poblenou.

In these areas, low accessibility is more closely related to urban form than to socioeconomic deprivation. Low-density residential patterns, separation from commercial streets, and peripheral or hillside geography reduce proximity to daily services.

For this reason, these neighbourhoods should not be the first target of redistributive public spending. Instead, accessibility gaps in wealthy but low-access areas can be addressed through planning incentives and mixed-use zoning. The municipality could simplify licensing procedures or provide targeted incentives for supermarkets, pharmacies, small medical clinics, and other daily services to open in suitable locations.

This approach recognizes the accessibility gap while preserving direct public investment for genuinely underserved low-income neighbourhoods.

#### **4.5 Recommendation 4: Protect High-Access Low-Income Neighbourhoods from Displacement**

The analysis also identifies neighbourhoods where low income and high accessibility coexist. El Raval is the clearest example, with a high accessibility score despite lower average income. This pattern shows that low-income neighbourhoods are not always underserved. In central Barcelona, historical density and mixed land use can produce strong service proximity.

However, this also creates a policy risk. High-accessibility low-income neighbourhoods may become attractive to higher-income residents, investors, and tourism-related real estate activity. If good accessibility increases housing demand and rent pressure, existing residents may be displaced from the areas that currently provide them with strong access to services.

For this reason, accessibility planning should be combined with anti-displacement policies. In high-accessibility low-income neighbourhoods such as el Raval and other central areas, Barcelona should strengthen affordable housing provision, tenant protection, social housing acquisition, and measures against speculative displacement. The objective should be to ensure that the benefits of the 15-minute city remain available to existing residents rather than being transferred only to wealthier incoming populations.

#### **4.6 Recommendation 5: Apply the Superblock Model Through an Equity Lens**

Barcelona's Superblock programme, or *Superilla*, is one of the city's most visible urban planning strategies. It aims to reduce car dominance, reclaim public space, and create more walkable and liveable neighbourhoods. However, future walkability interventions should be evaluated through an explicit equity lens.

The Low Access / Underserved cluster includes several neighbourhoods where street redesign alone would not be sufficient. In these areas, pedestrian improvements should be combined with the provision of local services. A Superblock intervention in an underserved area should not only reduce traffic, but

also create conditions for local markets, community facilities, green spaces, healthcare access, and safer routes to public transport.

Future Superblock projects should therefore be assessed according to two criteria. First, do they improve walkability and public space quality? Second, do they reduce accessibility inequality between neighbourhoods? This would help ensure that the programme contributes to spatial justice rather than only improving areas that are already well served.

#### **4.7 Recommendation 6: Establish an Annual 15-Minute City Monitoring Dashboard**

This study demonstrates that neighbourhood-level accessibility can be measured using open data and open-source tools. The methodology combines OpenStreetMap service data, municipal population and income data, GeoPandas spatial analysis, and a transparent accessibility index. Since the data sources are public and reproducible, the analysis could be updated regularly.

Barcelona should develop a public 15-minute city monitoring dashboard integrated with the Open Data BCN ecosystem. Rather than being real-time, the dashboard should be updated annually or whenever new official datasets become available. It should report accessibility scores for all seventy-three neighbourhoods, service counts by category, income and population indicators, and changes over time.

Such a dashboard would serve three purposes. First, it would allow policymakers to monitor whether accessibility gaps are narrowing or widening. Second, it would make smart city investments more transparent to residents. Third, it would provide researchers and civil society organizations with a common evidence base for evaluating urban policy.

The dashboard should also include methodological notes explaining the limitations of the indicator, especially the use of an 800-metre straight-line buffer rather than network-based walking time. This transparency is important because smart city indicators can influence investment decisions, and residents should understand how those indicators are constructed.

## **5 Ethical and Social Implications**

### **5.1 The Right to Urban Accessibility**

The findings of the research have significant ethical and social ramifications in addition to being technical. The 15-minute city cannot be assessed simply in terms of efficiency, sustainability, or smart technology if neighbourhoods have unequal access to basic services. It needs to be assessed as a spatial justice issue as well.

The concept of the right to the city, introduced by Lefebvre and later developed by Harvey, argues that urban residents should have meaningful access to the resources, services, and opportunities produced by the city [10, 11]. According to this viewpoint, having access to healthcare, education, grocery stores, public transportation, and public areas is more than just a convenience. It is a component of the residents' capacity to engage in urban life.

The analysis shows that this right is not equally supported across Barcelona's seventy-three neighbourhoods. The Low Access / Underserved cluster contains eighteen neighbourhoods with weaker service proximity and lower average household income. Based on the average population of this cluster, approximately 130,000 residents are affected by this double disadvantage. This does not mean that these residents have no access to services, but it does mean that they are less likely to benefit from the everyday proximity promised by the 15-minute city model.

### **5.2 Who May Be Excluded and Why**

The results suggest that exclusion from the 15-minute city can occur in different ways. The first and most policy-relevant group consists of residents living in low-income peripheral neighbourhoods,

especially in parts of Nou Barris, Horta-Guinardo, Sant Andreu, and the southern port zone. In these areas, lower income overlaps with weaker service proximity. This creates a double disadvantage: residents may have fewer economic resources while also facing longer distances to essential services.

A second group includes elderly residents, people with disabilities, and residents without access to private transport. Even where services are located within an 800-metre straight-line buffer, actual walking conditions may still be difficult. Steep terrain, unsafe crossings, poor lighting, inaccessible sidewalks, and fragmented pedestrian routes can reduce real accessibility. This is an important limitation of the present study because the indicator measures spatial proximity, not the full quality of the walking environment.

A third form of exclusion may affect residents who face social, linguistic, administrative, or cultural barriers when using services. A neighbourhood may have many services nearby, but physical proximity does not always guarantee effective access. For example, residents may still face difficulties related to affordability, language, opening hours, documentation, or trust in institutions. These dimensions are not directly measured in the accessibility score, but they are important for interpreting the results ethically.

### 5.3 The Role of the Data Scientist in Smart City Governance

This study also raises questions about the role of data scientists in smart city governance. Data analysis can support better decision-making, but it can also shape how urban problems are defined. The choice of indicators, weights, distance thresholds, and clustering methods influences which neighbourhoods appear well served or underserved.

For instance, the accessibility factor in this study gave healthcare facilities a higher weight than parks. This decision is an assessment of the relative significance of several services for day-to-day urban living. In the same way, although the study is visible and comparable due to the use of an 800-meter straight-line buffer, actual walking routes and obstacles in the pedestrian environment are not captured. These choices include normative presumptions in addition to technical ones.

Because of this, data scientists working in smart city environments need to be open about the methodological decisions they make. Indicators need to be transparent to the public, repeatable, and well-documented. Furthermore, locals in underprivileged areas must to be involved in the practical definition of accessibility. In order to guarantee that smart city indicators represent lived experience rather than just administrative or technical concerns, a participatory approach would be helpful.

The literature on public administration and automated decision-making warns that data-driven systems can reproduce social inequalities when they are designed without sufficient attention to vulnerable groups [12, 13]. In the context of the 15-minute city, this means that accessibility indicators should not only identify efficient service locations, but also support fairer and more inclusive urban policy.

### 5.4 Data Privacy and Ethical Data Use

The present study uses only aggregate neighbourhood-level data and publicly available geospatial datasets. It does not use individual-level personal information. For this reason, the privacy risks of the current analysis are limited.

Future developments in this approach, however, might give rise to more serious privacy issues. Strict data protection guidelines would be required, for instance, if real walking routes and travel patterns were estimated using mobile phone data, GPS traces, smart card data from public transport, or individual mobility records. Such information should be minimised, anonymised, and utilised exclusively for purposes that are clearly in the public interest.

An ethical smart city approach should therefore balance analytical precision with privacy protection. More detailed data can improve accessibility measurement, but it should not come at the cost of individual rights or surveillance risks. The objective of smart city analytics should be to improve public services and reduce inequality, not to monitor residents unnecessarily.

Overall, this study has a clear ethical implication: the 15-minute city should be considered more than just a technical design model. Ensuring that all neighbourhoods, particularly peripheral and lower-income regions, may benefit from being close to important services should be viewed as a social duty.

## 6 Conclusion

### 6.1 Summary of Findings

This study investigated if Barcelona can be considered as a 15-minute city that is equally accessible to all citizens. All seventy-three of Barcelona’s neighbourhoods, several publicly available datasets, and six categories of vital urban services—healthcare facilities, schools, supermarkets, pharmacies, parks, and metro stations—were the basis for the analysis.

The findings demonstrate that not every inhabitant of Barcelona can get there in 15 minutes. Accessibility shows a distinct spatial pattern and varies significantly between neighbourhoods. While outer, hillside, industrial, and lower-density regions typically have weaker service proximity, dense and mixed-use core areas typically do better.

Four main findings can be highlighted. First, the normalized accessibility scores range from 100.00 for la Vila de Gracia to 0.00 for Vallvidrera, el Tibidabo i les Planes. These values should be interpreted as relative scores within Barcelona, not as absolute measures of perfect or failed accessibility. The main pattern is a centre-periphery gradient, with stronger accessibility in central and inner-city neighbourhoods and weaker accessibility in several peripheral areas.

Second, the relationship between household income and accessibility is weak. The Pearson correlation coefficient between income and accessibility is 0.167, indicating that income alone is not a strong predictor of service proximity. Barcelona contains lower-income but highly accessible neighbourhoods, such as el Raval, and higher-income but less accessible neighbourhoods, such as Pedralbes. This shows that centrality, density, urban morphology, and land-use mix are also important factors.

Third, the K-Means clustering analysis identified four neighbourhood profiles: High Access, Medium-High Access, Medium-Low Access, and Low Access / Underserved. The Low Access / Underserved cluster contains eighteen neighbourhoods with weak service proximity and lower average income. Based on the average population of this cluster, approximately 130,000 residents are affected by this double disadvantage.

Fourth, the Medium-Low Access cluster shows that high income does not guarantee strong 15-minute city accessibility. Several wealthy western neighbourhoods have low accessibility scores because of their low-density residential character and weaker proximity to daily services. This confirms that the 15-minute city is not only a market outcome, but also a planning and governance challenge.

### 6.2 Contribution to Smart City Analysis

This study contributes to smart city analysis in three ways. First, it provides a reproducible methodology for measuring neighbourhood-level accessibility using public open data and open-source Python tools. The same approach could be adapted to other cities with comparable geospatial and socioeconomic data.

Second, the study combines spatial accessibility indicators with income and population data. This makes it possible to move beyond simple service counts and examine how accessibility overlaps with social inequality. The analysis shows that the most important equity issue is not income alone, but the overlap between low income and weak local service proximity.

Third, the cluster typology provides a practical framework for policy prioritization. Instead of treating all low-access neighbourhoods in the same way, the typology distinguishes between low-income underserved areas, wealthy but low-access areas, and high-access low-income areas at risk of displacement. Each group requires a different policy response.

### 6.3 Limitations and Future Research

This study has several limitations. The accessibility measure is based on an 800-metre straight-line buffer from the centroid of each neighbourhood. This provides a clear and comparable proximity indicator, but it does not measure actual walking routes, street network distance, slope, barriers, or pedestrian infrastructure quality. Future research should use network-based isochrone analysis with the walking street network downloaded through OSMnx to estimate more realistic walking times.

The service categories included in the analysis represent important daily needs, but they do not cover all dimensions of urban life. Future work could include childcare facilities, libraries, sports centres, cultural venues, post offices, and social services. Including these categories would provide a more complete picture of neighbourhood accessibility.

There is also a temporal limitation. Population and income data reflect the most recent available municipal datasets, while service locations were extracted from OpenStreetMap during the project period. Future versions of the analysis should use temporally consistent datasets where possible and update the accessibility scores annually.

Finally, the study measures physical proximity, not effective access. A service may be close but still difficult to use because of affordability, language barriers, opening hours, administrative requirements, or lack of trust. Future research should therefore combine spatial analysis with surveys, interviews, or fieldwork in underserved neighbourhoods.

### 6.4 Concluding Remarks

Barcelona is well known for its smart city projects and for its Superblock program, which emphasises walkability and proximity-based development. Still, this analysis demonstrates that the advantages of the 15-minute city model are not dispersed equally throughout the city. While certain remote and lower-income districts continue to have less access to basic amenities, central and dense neighbourhoods often perform well.

The policy implication is clear: Barcelona's smart city strategy should be guided by spatial equity. Public investment should prioritize neighbourhoods where low accessibility and lower income overlap. At the same time, high-access low-income areas should be protected from displacement, and wealthy but low-access areas should be improved through planning incentives and mixed-use development.

The six policy recommendations proposed in this report provide a roadmap for moving from a 15-minute city for some residents toward a 15-minute city for all residents. These recommendations include targeted public investment, improved public transport and last-mile connectivity, planning incentives, anti-displacement measures, an equity-oriented Superblock strategy, and an annual accessibility monitoring dashboard.

The broader lesson is that smart city analysis is not only a technical exercise. The choice of indicators, weights, thresholds, and visualizations shapes how urban problems are understood and which neighbourhoods receive attention. For this reason, the central question of smart city planning should remain: smart city for whom?

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